PROJECT 45: Design and Analysis of 9 Stage Ring Oscillator by using Cadence Virtuoso

A Comprehensive Study of Analog Circuits

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1. INTRODUCTION

A ring oscillator is a fundamental circuit widely used in digital and analog applications. It consists of a series of inverting stages connected in a closed-loop configuration, producing an oscillating output without requiring an external clock source. The primary applications of ring oscillators include frequency synthesis, timing circuits, clock generation, and as a test structure for process variations in semiconductor fabrication.

Ring oscillators are essential in VLSI (Very Large Scale Integration) circuits due to their simplicity and ease of integration. The frequency of oscillation depends on the number of stages and the propagation delay of each stage. In CMOS technology, ring oscillators are commonly implemented using an odd number of inverters to satisfy the Barkhausen criterion for oscillation.

This document presents the design and analysis of a **9-stage ring oscillator** using **Cadence Virtuoso**, a popular EDA (Electronic Design Automation) tool. Through this simulation environment, we can model the circuit, evaluate performance metrics, and analyze the effects of design parameters on oscillation frequency and stability.

2. BRIEF DESCRIPTION

A ring oscillator operates based on the delay introduced by each inverter stage. The circuit consists of **nine inverters** connected in a feedback loop, ensuring that the signal propagates continuously, generating a periodic waveform. The oscillation frequency is determined by:

where:

$$f = \frac{1}{2N \cdot t_d}$$

- is the number of inverter stages (9 in this case)
- is the delay per inverter stage

The oscillation occurs because each inverter introduces a delay, and the loop creates a phase shift that satisfies the oscillation conditions. Since each inverter provides a 180° phase shift and an odd number of inverters ensures an additional 180° phase shift due to feedback, the total phase shift of 360° ensures sustained oscillations.

3. CIRCUIT DESIGN

3.1 Schematic Design

The 9-stage ring oscillator is implemented using CMOS inverters. The key steps for schematic design in Cadence Virtuoso are:

1. **Create a new schematic cell view** in Cadence Virtuoso.
2. **Place 9 CMOS inverters** in a series configuration, ensuring that the output of one inverter is connected to the input of the next.
3. **Connect the last inverter's output back to the first inverter's input** to form a closed-loop structure.
4. **Define power supply connections (VDD and GND)** to provide the necessary operating conditions.
5. **Verify the schematic** using the Design Rule Check (DRC) and Electrical Rule Check (ERC) to ensure correctness.

3.2 Simulation Setup

After designing the schematic, simulation is performed to validate the circuit operation. The transient analysis setup includes:

1. **Selecting a simulation tool** such as Spectre in Cadence Virtuoso.
2. **Applying an initial condition** to one of the inverters to break symmetry and initiate oscillation.
3. **Defining simulation parameters**, such as:
    - **Time step** suitable for high-speed oscillations.
    - **Total simulation time** long enough to capture multiple oscillation cycles.
4. **Running transient analysis** to observe the oscillation waveform.
5. **Measuring oscillation frequency** using signal analysis tools within the Virtuoso Analog Design Environment (ADE).

3.3 Transient Response Analysis

The transient response provides insights into the oscillation behavior:

- **Waveform Shape:** The output signal should exhibit periodic transitions between high and low voltage levels.
- **Frequency Calculation:** The time period of one oscillation cycle is measured to determine the oscillation frequency.
- **Delay Per Stage:** The propagation delay of each inverter is derived from the total oscillation period.
- **Power Consumption:** The dynamic power consumption is evaluated based on switching activity and supply voltage.

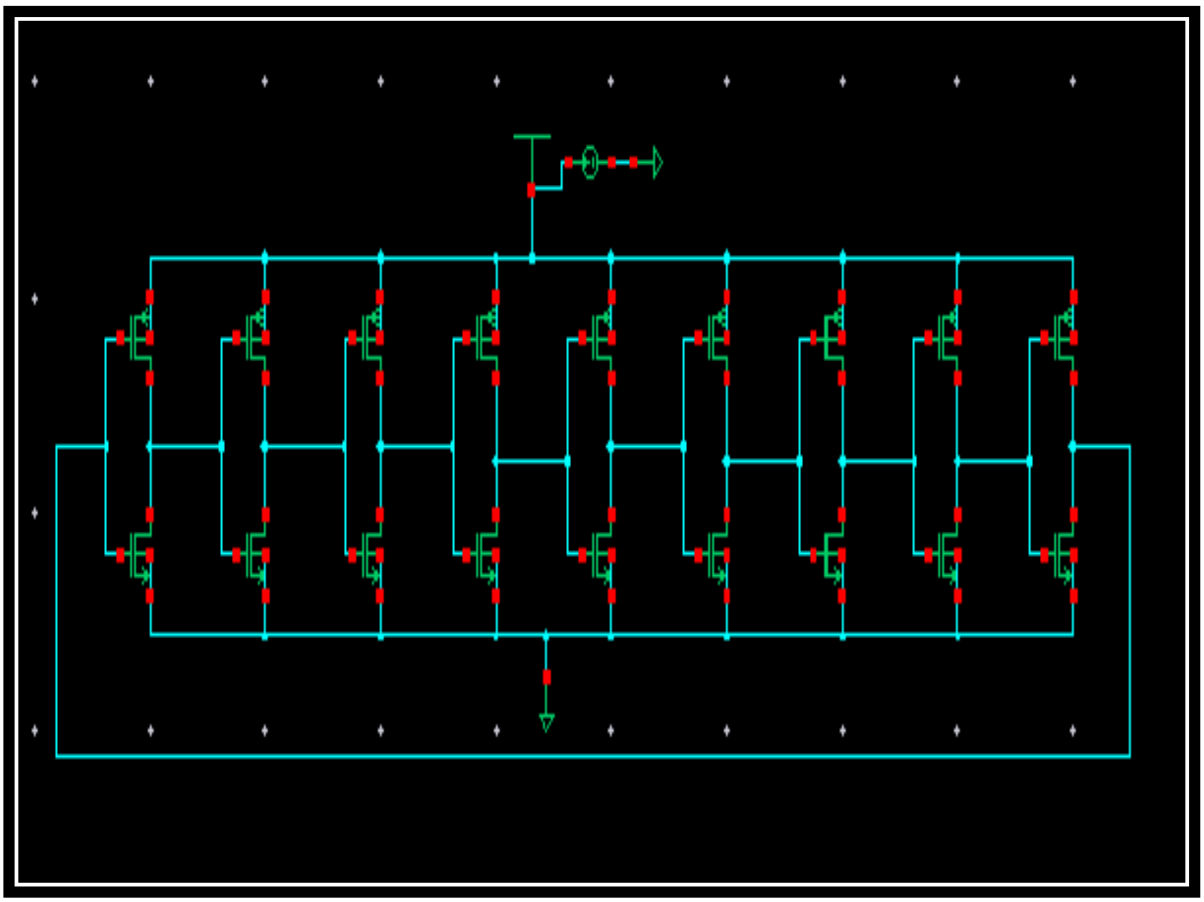

Figure 3.1 Circuit of 9 Stage Ring Oscillator

4. SIMULATION RESULTS

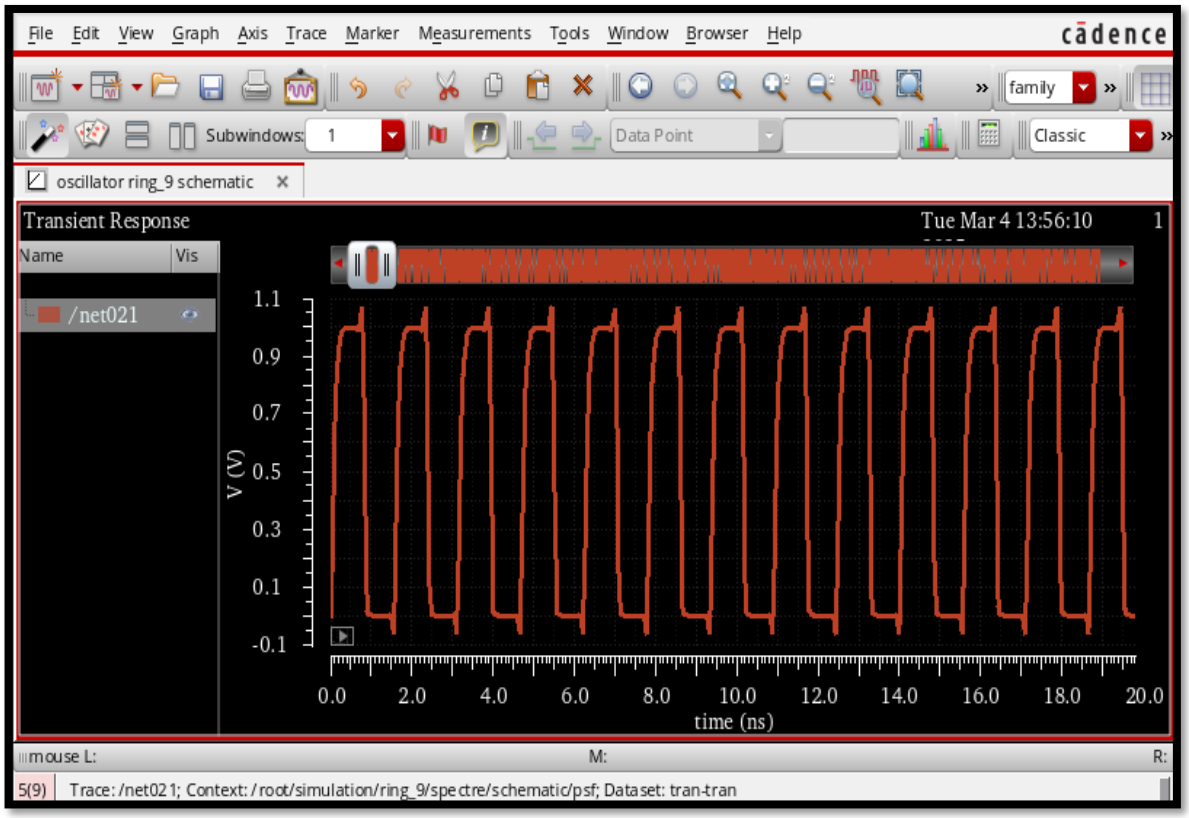

Figure 4.1: Simulation Results of 9 Stage Ring Oscillator

5. ADVANTAGES AND DISADVANTAGES

5.1 Advantages:-

1. **Simple Design:** The circuit requires only inverters and does not need an external clock.
2. **High Speed:** CMOS-based ring oscillators operate at high frequencies, making them useful for high-speed applications.
3. **Low Power Consumption:** CMOS technology ensures minimal static power dissipation.
4. **Integration:** Easily implementable in VLSI circuits, making them suitable for system-on-chip (SoC) designs.

5.2 Disadvantages:-

1. **Process Variations:** The oscillation frequency is highly sensitive to process variations, affecting consistency.
2. **Temperature Dependence:** Performance varies significantly with changes in temperature.
3. **Limited Frequency Stability:** Ring oscillators are susceptible to power supply noise and temperature fluctuations, making them less stable compared to crystal oscillators.

6. CONCLUSION

The 9-stage ring oscillator is a versatile and essential circuit widely used in modern electronic designs. It is particularly useful for generating high-speed clock signals, performing delay measurements, and testing semiconductor fabrication processes. The simplicity of its design makes it an attractive choice for many applications, while its drawbacks, such as frequency instability and process variations, must be carefully considered.

By using **Cadence Virtuoso**, we can effectively design, simulate, and analyze the behavior of the ring oscillator. Through transient analysis, we can determine the oscillation frequency, measure power consumption, and study the impact of transistor sizing and supply voltage on performance. The insights gained from this analysis enable engineers to optimize the design for specific applications, ensuring reliable and efficient operation.

Despite its limitations, the ring oscillator remains a valuable circuit in digital and analog design. Ongoing research continues to improve its performance, making it a critical component in modern semiconductor technology. With proper optimization, a 9-stage ring oscillator can be tailored for high-speed and low-power applications, making it an integral part of clock generation and signal processing circuits.

7. FREQUENTLY ASKED QUESTIONS (FAQs)

Q1. What is the significance of using an odd number of inverters in a ring oscillator?

Answer: An odd number of inverters ensures a total phase shift of 180° + 180° (due to the feedback), leading to sustained oscillations.

Q2. How can we control the frequency of a ring oscillator?

Answer: The frequency can be controlled by adjusting the number of inverter stages and modifying the inverter delay using transistor sizing or load capacitance.

Q3. Why is Cadence Virtuoso used for designing ring oscillators?

Answer: Cadence Virtuoso provides an industry-standard platform for designing and simulating analog and mixed-signal circuits with accurate modeling and process-specific parameters.

Q4. What is the impact of supply voltage on the oscillation frequency?

Answer: Higher supply voltage reduces inverter delay, increasing the oscillation frequency, while lower supply voltage decreases the frequency.

Q5. Can a ring oscillator be used as a random number generator?

Answer: Yes, due to its frequency instability and noise sensitivity, ring oscillators are often used in hardware random number generators (HRNGs).